

BCE-C611
SEMESTER EXAMINATION-MAY 2023
CLASS:B.TECH SEMESTER: VI
DISTRIBUTED SYSTEMS

Time: 3 hour

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A
(SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks.
(5 X 6 = 30 Marks)

- Question-1: What is a distributed file system and how does it differ from a traditional file system?
- Question-2: What is Domain Name and how it works in the distributed system. List the limitations of the Distributed systems.
- Question-3: What are some challenges that arise when designing distributed systems?
- Question-4: What is marshalling and why is it important in interprocess communication?
- Question-5: What is the role of the operating system layer in supporting distributed objects and remote invocation?
- Question-6: What is the role of name services in a distributed file system and how do they help to locate and access files?
- Question-7: What are some challenges involved in achieving distributed mutual exclusion?
- Question-8: What is the difference between physical time and logical time in a distributed system?
- Question-9: What is the difference between sequential consistency and release consistency in distributed shared memory systems?
- Question-10: How client server communication is possible in distributed systems?

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any *four* questions in detail. Each question carries 10 marks.

(4 X 10 = 40 Marks)

- Question-1:** Explain the architectural models of distributed systems, highlighting the advantages and disadvantages of each.
- Question-2:** Discuss design issues of DSM in detail.
- Question-3:** Discuss the authentication framework in Distributed System.
- Question-4:** Using the case study of Interprocess Communication in UNIX, describe the different types of interprocess communication mechanisms available in UNIX and their relative strengths and weaknesses.
- Question-5:** Describe the various mechanisms that operating systems use to support distributed objects and remote invocation. In your answer, discuss the role of the OS layer, protection, processes, threads, and communication, and how these mechanisms contribute to the overall OS architecture.
- Question-6:** Explain Distributed Mutual Exclusion and its types in detail.
- Question-7:** Discuss the concepts of clocks, events, and process states in a distributed system. Explain the role of physical clocks and the challenges involved in synchronizing them and describe how logical clocks can be used to establish a partial ordering of events. Using the topic of distributed debugging, explain how global states can be used to analyze the behavior of a distributed system.
- Question-8:** Explain the Common Object Request Broker Architecture (CORBA) and its role in distributed systems. Discuss the CORBA Remote Method Invocation (RMI) framework, including its design and implementation, and the services that it provides. Using a case study, discuss how CORBA is used in practice to build distributed systems.

No. of Copies: 32

First Sessional Examination

Time: 1 Hr.

Date: 01.03.2024

Distributed Systems (BCE-C611)

Maximum Marks: 20

Note: Attempt any 2 questions from (Q1-Q3) and each question carries 5 marks
Question Q4 is compulsory for all and question carries 10 marks.

Q1 Write a short note on Distributed Computing and its characteristics? (5) (Understand, L-01, CO-01)

Q2 What are the different challenges in distributed systems. (5) (Knowledge, L-01, CO-01)

Q3 Explain client server communication and RPC with diagram. (5) (Understand, L-01, CO-01)

Q4 What are the different types of distributed systems and explain its advantages and disadvantages? (10) (Knowledge, L-01, CO-02)

****All the Best****

BCE-C611
SEMESTER EXAMINATION- MAY 2024
CLASS: B.TECH. SEMESTER: 6TH
DISTRIBUTED SYSTEMS

Max. Marks: 70

Time: 3 hours

Note: Question Paper is divided into two sections: A and B. Attempt both the sections as per given instructions.

CO BL

SECTION-A**(SHORT ANSWER TYPE QUESTIONS)**

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks. (5× 6 = 30 Marks)

- | | | |
|---|---|---|
| 1. List the limitations of the Distributed systems. | 1 | 5 |
| 2. Explain Sun NFS. | 3 | 3 |
| 3. Explain Terms: Event, Process, Threads, Operating System, Client-server. | 2 | 2 |
| 4. How client server communication is possible in distributed systems. | 1 | 4 |
| 5. Discuss the different API are used in Internet. | 1 | 5 |
| 6. Explain the synchronization with physical and logical clocks. | 2 | 6 |
| 7. Explain domain names and web servers with examples. | 2 | 3 |
| 8. Explain external data representation and marshalling concepts in detail. | 4 | 5 |
| 9. Explain Process states in detail with diagram. | 2 | 1 |
| 10. Explain in details the various components of RMI in the java with suitable example. | 3 | 1 |

SECTION-B

(LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any *four* questions in detail. Each question carries 10 marks.

(4 × 10 = Marks)

- | | | | |
|----|---|---|---|
| 1. | Explain the different consistency models for distributed. | 3 | 5 |
| 2. | Describe the various RPC protocols supporting client server communication. | 2 | 1 |
| 3. | Describe CORBA RMI and its services. | 5 | 5 |
| 4. | Differentiate TCP stream communication and Client Server Communication. | 3 | 2 |
| 5. | What is name service? What are its goals? How is it implemented? What is directory service? | 4 | 3 |
| 6. | Explain the different consistency models for distributed. ✕ | 3 | 6 |
| 7. | Compare and contrast between DNS and GNS with their advantages and disadvantages. | 5 | 5 |
| 8. | Explain the following terms in details: | 1 | 4 |
| | a) Global start | | |
| | b) Mutual Exclusion | | |
| | c) Logical Exclusion | | |

PAPER CODE: BCE-C611
SEMESTER EXAMINATION (MAY-2025)
CLASS: B.TECH
SEMESTER: VI
SUBJECT: CSE
PAPER TITLE: DISTRIBUTED SYSTEMS

Time: 3 Hours

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)		CO	BL
Instructions: Answer any <i>five</i> questions in about 150 words each. Each question carries six marks. (5 x 6 = 30 Marks)			
Q-1	Explain about Distributed Systems? Explain their key characteristics with examples.	CO1	BL1
Q-2	Define Events and Notifications in a distributed system? Explain with examples.	CO2	BL2
Q-3	Explain the concept of Distributed Shared Memory (DSM) in distributed systems. What are the key design and implementation challenges associated with DSM?	CO1	BL1
Q-4	Explain the terms: 1) Replication 2) Consistency 3) Concurrency.	CO3	BL3
Q-5	Define Distributed Objects and Remote Invocation? Explain their importance in distributed systems.	CO2	BL1
Q-6	Explain the role of the Operating System in Distributed Computing?	CO2	BL2
Q-7	Explain about: 1) Race Condition 2) Context Switching 3) SOA.	CO4	BL2
Q-8	Explain the significance of global states in distributed systems and their role in debugging, deadlock detection, and fault tolerance.	CO1	BL2
Q-9	Provide a case study on Interprocess Communication (IPC) in UNIX, comparing pipes, message queues, shared memory, and sockets, including security and efficiency trade-offs.	CO4	BL4
Q-10	Discuss the Common Object Request Broker Architecture (CORBA) and its role in distributed computing.	CO3	BL3
SECTION-B (LONG ANSWER TYPE QUESTIONS)		CO	BL
Instructions: Answer any <i>four</i> questions in detail. Each question carries 10 marks. (4 x 10 = Marks)			
Q-1	Discuss the concept of resource sharing in distributed systems, focusing on the Web. How do middleware, caching, and load balancing contribute to efficient resource management?	CO3	BL4
Q-2	Explain the system models used in distributed systems. Discuss the differences between three tier models and peer-to-peer models with examples.	CO2	BL2

Q-3	Discuss algorithms for distributed mutual exclusion and leader election, comparing Ricart-Agrawala, Maekawa's algorithms with their advantages and drawbacks.	CO3	BL2
✓ Q-4	Explain the role of events, and process states in distributed systems and their impact on consistency and coordination through time synchronization.	CO2	BL3
<u>Q-5</u>	Explain the role of Name Services in distributed computing and how DNS ensures fault tolerance and load balancing in large-scale networks	CO1	BL1
Q-6	Discuss File Service Architecture in distributed systems, comparing stateful and stateless file services in terms of performance, reliability, and scalability.	CO4	BL2
✓ Q-7	Explain the concept of marshalling in distributed systems. How does it facilitate communication between heterogeneous systems, and what are the challenges associated with marshalling and unmarshalling data?	CO1	BL1
✓ Q-8	Discuss the importance of physical clock synchronization in distributed systems and compare Cristian's algorithm, Berkeley's algorithm, and NTP.	CO2	BL3

1st SESSIONAL EXAMINATION-2025
CLASS – VIth SEM B. TECH (CSE)
SUBJECT- Distributed Systems (BCE-C621)

Date-23-02-2026
Time-2.00PM to 3.00PM

Section A

Answer any TWO questions. Each question carries 6 marks.

Q1	Explain the concept of a distributed system. Discuss its importance and provide real-world examples	CO1	BL-1
Q2	Explain the concept of Interprocess Communication (IPC) in distributed system	CO2	BL-2
Q3	Explain about Coupling and its types in details	CO3	BL-3
Q4	Explain about process and threads in details	CO1	BL-1

Section B

Answer any ONE questions. Each question carries 8 marks.

Q1	Explain the system models used in distributed systems. Discuss the differences between client-server and peer-to-peer models with examples.	CO2	BL-2
Q2	Compare TCP and UDP protocols in distributed systems with respect to connection establishment, reliability, flow control, and suitable application scenarios	CO3	BL-3

Distributed Systems(BCE-C621)

Date: 20.04.26

Time: 2.00-3.00

Section A

Answer any TWO questions. Each question carries 6 marks.

Q1	Explain about Distributed File System (DFS), its advantages and limitations.	BL1	CO1
Q2	Explain about File permissions and their types.	BL3	CO2
Q3	Explain about clock and their types in details	BL2	CO3
Q4	Define: i) Mutual Exclusion 2) Critical Section 3) Race Condition	BL1	CO2

Section B

Answer any ONE questions. Each question carries 8 marks

Q1	Explain clock synchronization in distributed systems. Discuss different techniques such as NTP and logical clocks with their advantages and limitations.	BL1	CO1
Q2	What is deadlock in distributed systems? Explain its causes, detection methods, and prevention techniques.	BL3	CO2

PAPER CODE: BCE-C621
EVEN SEMESTER EXAMINATION (MAY-2026)
CLASS: B.TECH. SEMESTER: VI
PAPER TITLE: DISTRIBUTED SYSTEMS

Time: 3 hours

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks.
 (5 x 6 = 30 Marks)

		CO	BL
Q-1	Discuss the role, functions, and importance of the Operating System in Distributed Computing systems	CO2	BL2
Q-2	Define events and notifications in distributed systems and discuss their roles with appropriate real-world examples.	CO2	BL4
Q-3	Explain the role of APIs in Inter-process communication.	CO1	BL1
Q-4	Explain client-server architecture with examples and diagrams.	CO3	BL3
Q-5	Explain the terms: 1) Replication. 2) Consistency. 3) Concurrency.	CO3	BL3
Q-6	Explain how the HTTP protocol works in web communication	CO1	BL1
Q-7	Explain about: 1) Race Condition. 2) Context Switching. 3) SOA.	CO4	BL2
Q-8	What is the purpose of logical clocks in distributed systems? How do they help in event ordering?	CO1	BL1
Q-9	How does CORBA facilitate communication among distributed objects?	CO4	BL4
Q-10	Briefly explain election algorithms and multicast communication	CO2	BL2

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any *four* questions in detail. Each question carries 10 marks.
 (4 x 10 = 40 Marks)

		CO	BL
Q-1	Explain the concept of marshalling in distributed systems and discuss the challenges faced in different (heterogeneous) environments.	CO3	BL4
Q-2	Describe different system models in distributed systems and compare peer-to-peer and three-tier architectures with suitable examples.	CO2	BL2

Q-3 -	Explain how DNS works as a name service and how it helps in fault tolerance and load balancing.	CO3	BL2
Q-4	Compare stateful and stateless file services in distributed systems with respect to performance, reliability, and scalability.	CO2	BL3
Q-5	Explain the role of events and process states in distributed systems and how they help in coordination and maintaining consistency.	CO1	BL1
Q-6 -	Discuss how caching, middleware, and load balancing improve the performance of web-based distributed systems.	CO4	BL2
Q-7	Compare Ricart-Agarwala and Maekawa algorithms for distributed mutual exclusion with their advantages and limitations.	CO1	BL1
Q-8 -	Explain different clock synchronization techniques such as Cristian's, Berkeley, and NTP, and compare their working.	CO2	BL3
